if α_i is a simple root, then V_{α_i} has a 1-dimensional space at i and is 0 everywhere else.

(a) Show that if i is a source, then $\mathrm{Ext}^1(V, V_{\alpha_i}) = 0$ for any representation V of Q, and if i is a sink, then $\mathrm{Ext}^1(V_{\alpha_i}, V) = 0$.

(b) Given an orientation of the quiver, find a Jordan-Hölder series of V_α for that orientation.